THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 10: **Light: Mirrors and Lenses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A clock's image in a plane mirror shows 10:10 in a mirror. The actual time is: 1

(a) 2:50 **(b)** 1:50 **(c)** 11:50 **(d)** 10:10

2. If the object distance from a convex mirror is increased, the image becomes: 1

(a) inverted **(b)** real

(c) smaller and closer to F **(d)** larger

3. Assertion (A): The focal length of a plane mirror is infinite. Reason (R): Rays parallel to the axis do not converge after reflection from a plane mirror. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

4. An object is moved from 2f to f of a concave mirror along the axis. The image moves from: 1

(a) C to F **(b)** F to P **(c)** C to infinity **(d)** infinity to C

5. A beam of light passes undeviated through a lens at: 1

(a) its optical centre **(b)** any point

(c) its focus **(d)** its edge

6. Assertion (A): A concave lens is used for correcting myopia. Reason (R): A concave lens converges light rays. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

7. A concave mirror forms a real inverted image three times larger than the object. The object is placed: 1

(a) at C **(b)** between C and F **(c)** beyond C **(d)** between F and P

8. A converging lens with a short focal length is a more powerful lens because it: 1

(a) is lighter **(b)** bends them less

(c) does not bend **(d)** bends rays more strongly

9. Assertion (A): We can see a book in a room. Reason (R): The book reflects light that falls on it into our eyes. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. A convex lens is cut along its principal axis into two halves. Each half will: 1

(a) become concave lenses
(b) become mirrors
(c) not form images
(d) form an image of the same type but dimmer

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Name the nature of a lens that can give both enlarged real and virtual images. State the conditions. 2

12. A mirror shows a letter 'b' appearing as 'd'. Explain. 2

13. Why are surgeons' and dentists' mirrors made small and concave? 2

14. State why a diverging lens is placed in peepholes of doors. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how a periscope's two mirrors are arranged and how the final image seen is erect, virtual and same size. 3

16. How can you identify without touching whether a lens is convex or concave using sunlight? 3

17. A ray of light passing through the centre of curvature of a mirror is reflected back along its path. Explain. 3

OR

Compare the images formed by a concave mirror and a convex lens for an object beyond twice the focal length. 3

18. A convex lens of focal length 25 cm forms an image of the Sun. How far must the screen be from the lens and what is the nature of the image? 3

OR

Explain how a single plane mirror can be used to measure the height of a tree using a simple method with shadows or with a stick. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Discuss how mirrors and lenses are applied in telescopes, microscopes, cameras and projectors, explaining the type of lens/mirror and the nature of the image. 5

OR

Explain in steps how to draw the ray diagram for an object placed between the pole and the focus of a concave mirror and state the nature of the image. Why is such a mirror used by barbers? 5

20. Explain lateral inversion and write what a person sees in a mirror for the word 'MAHESH', the number '2018' and an analogue clock showing 4:30. 5

OR

Investigate the relation between the number of images and the angle between two plane mirrors by planning an experiment. Provide a table for 30, 45, 60, 90, 120 degrees and the formula. 5

END OF QUESTION PAPER Total: 40 Marks

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